

Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Date: Aug 31, 2023

Pre-Tutorial Assignment 4

Marks: 2

1. Consider the electric circuit shown in Figure 1. The circuit elements have the following numerical values – $R_1 = 20\Omega$, $R_2 = 10\Omega$, $R_3 = 30\Omega$, $I_1 = 4A$, $V_1 = 60V$, $I_d = k \times i = 0.4 \times i$. Evaluate the voltage V_{R3} across the resistor R_3 using the *Superposition Theorem*. Report the numerical values of $V_{R3}(I_1)$ (when only I_1 is active) and $V_{R3}(V_1)$ (when only V_1 is active).

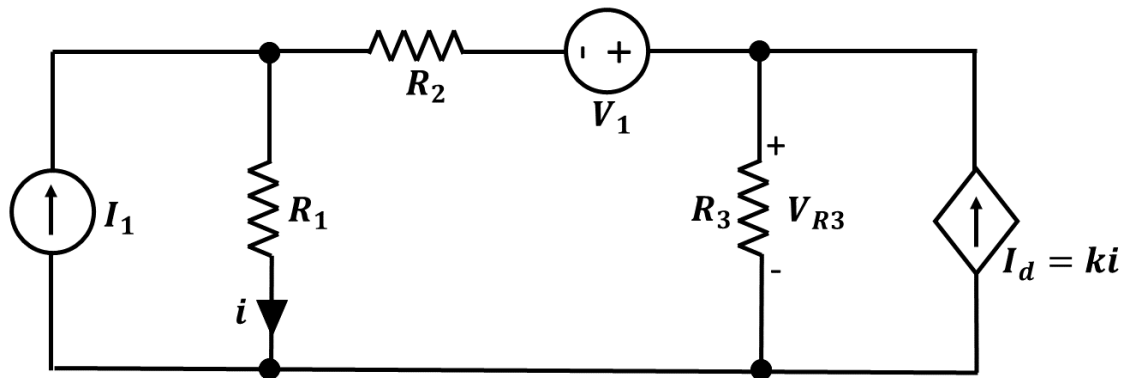


Figure 1: Electric circuit with two independent sources and one dependent source.

2. Consider the Boolean function

$$F(A, B, C) = \Sigma(0, 1, 2, 5, 6, 7).$$

- (a) Draw the K-map for the function F .
- (b) Find all Boolean expressions for F in the minimal SOP form.
- (c) Find all Boolean expressions for F in the minimal POS form.